This is the magnification for normal focusing.

The magnification for normal focusing is one less than that of near point focusing. But, the viewing is more comfortable in normal focusing than near point focusing. For large values of D/f, the difference between the two magnifications is negligibly small.

EXAMPLE 7.19

A man with a near point of 25 cm reads a book which has small print using a magnifying lens of focal length 5 cm. (a) What are the closest and the farthest distances at which he should keep the lens from the book? (b) What are the maximum and the minimum magnification possible?

Solution

$D = 25$ cm

The magnifying lens must be a convex lens of positive focal length $f = 5$ cm

For closest object distance u, the image distance v is, –25 cm. (near point, $v = -D$)

For farthest object distance u', the corresponding image distance, v' is infinity.

(a) To find closest distance between lens and book, we can use lens equation,

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

Rewriting for closest object distance,

$$\frac{1}{u}=\frac{1}{v}-\frac{1}{f}$$

Substituting,

$$\frac{1}{u}=\frac{1}{-25}-\frac{1}{5}=\frac{1}{25}-\frac{1}{5}=\left(\frac{-1-5}{25}\right)=-\frac{6}{25}$$

$$u=-\frac{25}{6}=-4.167 \text{ cm}$$

The closest distance between the lens and the book is, $u = -4.167$ cm

To find farthest object distance, lens equation is, $\frac{1}{v'}-\frac{1}{u'}=\frac{1}{f'}$

Rewriting for farthest object distance,

$$\frac{1}{u'}=\frac{1}{v'}-\frac{1}{f'}$$

Substituting, $\frac{1}{u'}=\frac{1}{\infty}-\frac{1}{5};u=-5$ cm

The farthest distance at which the person can keep the book is, $u' = -5$ cm

(b) To find magnification in near point focusing, $m=1+\frac{D}{f}=1+\frac{25}{5}=6$

To find magnification in normal focusing,

$$m=\frac{D}{f}=\frac{25}{5}=5$$

7.6.1.3. Resolving power of microscope

A microscope is used to see the details of the object under observation. Good microscope should not only magnify the object but also resolve the two points on an object which are separated by the smallest distance d_{min}. Actually, d_{min} is the resolution and its reciprocal is the resolving power.

The spatial resolution (radius of central maximum) is already derived in equation (7.54), $r_0=\frac{1.22\lambda f}{a}$.

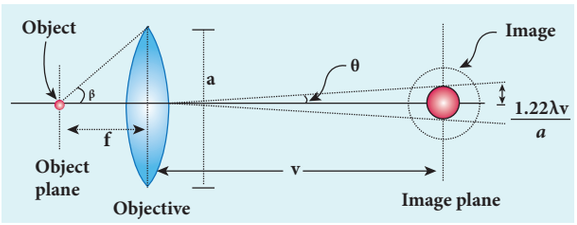


Figure 7.38 Resolving power of microscope

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